

Jessica Bavaresco

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Position

Feb 2025 – **Research faculty member (*chargée de recherche CNRS*)**
present **Quantum Information (QI) team, LIP6**
CNRS & Sorbonne University
Paris, France

Education

Mar 2017 – **PhD in Physics** University of Vienna, Vienna, Austria
Jun 2021 Thesis: *Certifying complex quantum properties: High-dimensional entanglement and indefinite causal order*
Graduated with honors, on 21 June 2021.
Supervisor: Prof. Marcus Huber
Mar – Jun **PhD Internship** The University of Tokyo, Tokyo, Japan
2019 Supervisor: Prof. Mio Murao
Aug 2014 – **Master in Physics** Federal University of Minas Gerais, Belo Horizonte, Brazil
Aug 2016 Thesis: *When Bob cannot trust Alice: A semi-device-independent tale of quantum steering*
Supervisor: Prof. Marcelo Terra Cunha
Mar 2010 – **Bachelor in Physics** Federal University of Santa Catarina, Florianópolis, Brazil
July 2014 Jan – Dec **Bachelor exchange programme** Lafayette College, Easton-PA, USA
2012 Supervisor: Prof. Anthony D. Novaco

Previous positions

Oct 2023 – **Senior postdoctoral researcher (*maître assistante*)** – University of Geneva, **Switzerland**
Jan 2025 Supervisor: Prof. Nicolas Brunner
Funding: SNSF Swiss Postdoctoral Fellowship (Switzerland)
Jun 2022 – **Postdoctoral researcher** – University of Geneva, **Switzerland**
Sep 2023 Supervisor: Prof. Nicolas Brunner
Funding: NCCR SwissMAP (Switzerland)
Jul 2021 – **Postdoctoral researcher** – IQOQI Vienna, ÖAW, **Austria**
May 2022 Young Independent Research Group (YIRG)
Supervisors: Dr. Ämin Baumeler, Dr. Costantino Budroni, and Dr. Yelena Guryanova
Funding: FWF-Zukunftskolleg (Austria)
Mar – Jun **Visiting PhD student** – The University of Tokyo, **Japan**
2019 Supervisor: Prof. Mio Murao
Funding: FWF-START Prize (Austria)

Oct 2016 – **PhD student** – IQOQI Vienna, ÖAW, **Austria**
Jun 2021 Supervisor: Prof. Marcus Huber
Funding: FWF-START Prize (Austria)

Fellowships and grants

Research grants

Jul 2026 **AAPG 2026 Jeunes Chercheuses et Jeunes Chercheurs (JCJC)**
Agence Nationale de la Recherche – ANR (France): **368k EUR**
PI of Project: *Simulating Higher-Order Quantum Computing: From Causal Structures to Quantum Circuit Implementations (HIGHER)*

Oct 2017 **ESQ Discovery Grant 2017**
Austrian Academy of Sciences – ÖAW (Austria): **15k EUR**
PI of Project: *Mutually unbiased bases – the notorious case of dimension 6.*
Research grant that supports innovative and high-risk projects.

Postdoctoral fellowships

Jun 2023 **SNSF Swiss Postdoctoral Fellowship 2022 (“Swiss MSCA”)**
Swiss National Science Fund – SNSF (Switzerland): 249k CHF ~ **260k EUR** (in 2023)
PI of Project: *Beyond Entangled Pairs: Quantum Correlations in Networks (BEPQCiN)*

Feb 2023 **MSCA Individual Postdoc Fellowship 2022 (Global Fellowship)**
Score: 96.2% — Top 8.77% of candidates
European Research Executive Agency (EU): **309k EUR**
PI of Project: *Beyond Entangled Pairs: Quantum Correlations in Networks (BEPQCiN)*

Student grants

Jun 2014 **Federal Master’s Research Stipend**
CAPES (Brazil): 36k BRL ~ **11k EUR** (in 2014)
Two-year Master’s level research stipend awarded through the selection process of UFMG, Brazil

Sep 2011 **Science Without Borders Scholarship**
CAPES (Brazil): ca. 100k BRL ~ **44k EUR** (in 2011)
Excellence-based fully funded one-year abroad Bachelor’s level scholarship at Lafayette College, USA, awarded via a national selection process.

Editorial and conference service

Editorial boards

Jan 2026 – **Editorial Board of PRA** – Member of the Editorial Board of the journal Physical Review A of the
present American Physical Society (APS).

Conference committees

Sep 2026 – Nov 2026	Programme Committee for QIP 2027 – 30th Annual Quantum Information Processing Conference.
Jul 2026 – Dec 2026	Organizing Committee for JFQI 2026 – 2nd Japanese-French Quantum Information Workshop. Tokyo, Japan.
Jun 2023 – present	Steering Committee for YQIS – International Conference for Young Quantum Information Scientists.
Feb – Mar 2025	Programme Committee for QPL 2025 – 22nd International Conference on Quantum Physics and Logic.
Oct 2017 – Sep 2018	Organizing Committee for YQIS-CoQuS 2018 – 4th International Conference for Young Quantum Information Scientists & Summer School of the Vienna Doctoral Program in Complex Quantum Systems. Vienna, Austria.

Supervision

PhD students

Nov 2025 – present	Camiel Meijer . <i>Fundamental limits of signalling in quantum theory</i> , Sorbonne University, France.
Jan 2025 – present	Zoe García del Toro . <i>Higher-order quantum computing and indefinite causal order</i> , Sorbonne University, France.

Master's students

Apr – Sep 2025	Camiel Meijer , <i>Towards an information-theoretic principle to delimit quantum communication scenarios</i> , Sorbonne University, France.
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Bachelor's students

Feb – Mar 2026	Haruka Kanazawa , Intern student from the University of Tokyo, Japan, <i>Universal quantum gate reordering</i> , Sorbonne University, France.
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Evaluation and reviewing activities

Thesis examination jury

Aug 2026	PhD thesis reviewer – Edwin Peter Lobo, “ <i>Quantum Correlations: From Full Nonlocality to Device-Independent Certification</i> ”. Supervised by Prof. Stefano Pironio. PhD in Physics, Université libre de Bruxelles, Belgium.
Jun 2026	Master's thesis reviewer – Sabrina Camargo Zani, “ <i>Indefinite causality and quantum mechanics</i> ”. Supervised by Prof. Marcelo Terra Cunha. Master's in Applied Mathematics, State University of Campinas (Unicamp), Brazil.

- Feb 2025 **Master's thesis reviewer** – Gustavo Balvedi Pimentel, “*Symmetries of non-contextuality inequalities and the statistical robustness of experimental implementations*”. Supervised by Prof. Bárbara Lopes Amaral. Master's in Physics, University of São Paulo (USP), Brazil.
- Nov 2023 **PhD thesis reviewer** – Timothée Hoffreumon, “*Characterization of Higher-order Quantum Processes: When projective methods recover a model of logic*”. Supervised by Prof. Ognyan Oreshkov. PhD in Engineering Sciences and Technology, Université libre de Bruxelles, Belgium.

Thesis monitoring jury

- Feb 2026 & Sep 2026 **PhD monitoring committee (CSI), external member** – Kieran McShane, “*Moment-based PPT of random bipartite quantum states*”, years 1 & 2. Supervised by Dr. Cécilia Lancien and Dr. Ion Nechita. Université Grenoble Alpes, France.
- Jul 2025 **Master's project qualification, external member** – Sabrina Camargo Zani, “*Causalidade indefinida e mecânica quântica*”. Supervised by Prof. Marcelo Terra Cunha. State University of Campinas (Unicamp), Brazil.

Reviewer

- 2017 – present **Journals:** Physical Review Letters, PRX Quantum, Physical Review A, New Journal of Physics, Quantum Journal, Quantum Information Processing, and European Physics Journal Plus.
- 2022 – present **Conferences:** TQC 2022, TQC 2024, TQC 2025, QIP 2024, QIP 2025, QIP 2026, and Causal-worlds 2024.

Teaching

- Autumn 2025 **M2 Quantum Information Theory**, lectures (4h) and tutorials (4h). Computer Science and Physics Master's in Quantum Information, Sorbonne University, France.
- Spring 2025 **M1 Projet Moyen** (Long Project), project supervision (4h). Computer Science Master's in Quantum Information, Sorbonne University, France.
- Spring 2023 **L1 Méthodes Mathématiques pour Physiciens I** (Mathematical Methods for Physicists I), tutorials (16 hours). Physics Bachelor's course, University of Geneva, Switzerland.
- Autumn 2022 **L1 Méthodes Mathématiques pour Physiciens I** (Mathematical Methods for Physicists I), tutorials (16 hours). Physics Bachelor's course, University of Geneva, Switzerland.

Publications and research output

All of my research articles can be found on [my arXiv page](#). A full list of publications is also available on [my Google Scholar page](#), which counts over **1.1k citations** (Sep 2026).

Pre-prints

- [P1] Purification enables an unbounded query separation in sequential quantum channel discrimination
Z.G. del Toro, M.T. Quintino, and **J. Bavaresco**
[arXiv:2609.40218 \[quant-ph\]](#) (2026)

- [P2] The minimal example of quantum network Bell nonlocality
E. Don, **J. Bavaresco**, P. Lipka-Bartosik, N. Gisin, N. Brunner, and A. Pozas-Kerstjens
[arXiv:2605.00981 \[quant-ph\]](#) (2026)
- [P3] Probabilistic and approximate universal quantum purification machines
Z.G. del Toro and **J. Bavaresco**
[arXiv:2604.06325 \[quant-ph\]](#) (2026)
- [P4] Indefinite causal order in boxworld theories
J. Bavaresco, Ä. Baumeler, Y. Guryanova, and C. Budroni
[arXiv:2411.00951 \[quant-ph\]](#) (2024)

Published

- [P5] Strategy optimization for Bayesian quantum parameter estimation with finite copies: Adaptive greedy, parallel, sequential, and general strategies
E.L. André, **J. Bavaresco**, and M. Mehboudi
Quantum Science and Technology **11**, 035040 (2026)
[arXiv:2602.09655 \[quant-ph\]](#)
- [P6] The power of quantum catalytic local operations
P. Lipka-Bartosik, **J. Bavaresco**, N. Brunner, and P. Sekatski
Physical Review A **113**, 062216 (2026)
[arXiv:2510.08320 \[quant-ph\]](#)
- [P7] Simulating the quantum switch with quantum circuits is computationally hard
J. Bavaresco, H. Kristjánsson, M. Murao, T. Otake, M.T. Quintino, P. Taranto, S. Yoshida
Nature Communications **16**, 10216 (2025)
[arXiv:2409.18420v2 \[quant-ph\]](#)
Obs.: combines the results of [arXiv:2409.18420v1 \[quant-ph\]](#) and [arXiv:2409.18202 \[quant-ph\]](#)
- [P8] Catalytic Activation of Bell Nonlocality
J. Bavaresco, N. Brunner, A. Girardin, P. Lipka-Bartosik, and P. Sekatski
Physical Review Letters **135**, 220203 (2025)
[arXiv:2504.02042 \[quant-ph\]](#)
- [P9] Designing optimal protocols in Bayesian quantum parameter estimation with higher-order operations
J. Bavaresco, P. Lipka-Bartosik, P. Sekatski, and M. Mehboudi
Physical Review Research **6**, 023305 (2024)
[arXiv:2311.01513 \[quant-ph\]](#)
- [P10] Semi-device-independent certification of indefinite causal order in a photonic quantum switch
H. Cao, ^{*}**J. Bavaresco**, ^{*}N.-N. Wang, L.A. Rozema, C. Zhang, Y.-F. Huang, B.-H. Liu, C.-F. Li, G.-C. Guo, and P. Walther
^{*}These authors contributed equally to this work.
Optica **10**, 561 (2023)
[arXiv:2202.05346 \[quant-ph\]](#) (2022)
- [P11] Resource theory of causal connection
S. Milz, **J. Bavaresco**, and G. Chiribella

- Quantum **6**, 788 (2022)
arXiv:2110.03233 [quant-ph]
- [P12] Activation of genuine multipartite entanglement: Beyond the single-copy paradigm of entanglement characterisation
H. Yamasaki, S. Morelli, M. Miethlinger, **J. Bavaresco**, N. Friis, and M. Huber
Quantum **6**, 695 (2022)
arXiv:2106.01372 [quant-ph]
- [P13] Unitary channel discrimination beyond group structures: Advantages of sequential and indefinite-causal-order strategies
J. Bavaresco, M. Murao, and M. T. Quintino
Journal of Mathematical Physics **63**, 042203 (2022)
arXiv:2105.13369 [quant-ph] (2021)
- [P14] Strict hierarchy between parallel, sequential, and indefinite-causal-order strategies for channel discrimination
J. Bavaresco, M. Murao, and M. T. Quintino
Physical Review Letters **127**, 200504 (2021)
arXiv:2011.08300 [quant-ph]
- [P15] Verification of high-dimensional entanglement generated in quantum interference
Y. Chen, S. Ecker, **J. Bavaresco**, T. Scheidl, L. Chen, F. Steinlechner, M. Huber, and R. Ursin
Physical Review A **101**, 032302 (2020)
arXiv:1910.07684 [quant-ph]
- [P16] Semi-device-independent certification of indefinite causal order
J. Bavaresco, M. Araújo, Č. Brukner, and M. T. Quintino
Quantum **3**, 176 (2019)
arXiv:1903.10526 [quant-ph]
- [P17] Measurements in two bases are sufficient for certifying high-dimensional entanglement
J. Bavaresco, N. H. Valencia, C. Klöckl, M. Pivoluska, P. Erker, N. Friis, M. Malik, and M. Huber
Nature Physics **14**, 1032 (2018)
arXiv:1709.07344 [quant-ph]
- [P18] Simulations of submonolayer Xe on Pt(111): The case for a chaotic low temperature phase
A. D. Novaco and **J. Bavaresco**
Journal of Chemical Physics **148**, 144704 (2018)
arXiv:1708.01493 [cond-mat.mtrl-sci]
- [P19] Operational framework for quantum measurement simulability
L. Guerini, **J. Bavaresco**, M. Terra Cunha, and A. Acín
Journal of Mathematical Physics **58**, 092102 (2017)
arXiv:1705.06343 [quant-ph]
- [P20] Most incompatible measurements for robust steering tests
J. Bavaresco, M. T. Quintino, L. Guerini, T. O. Maciel, D. Cavalcanti, and M. Terra Cunha
Physical Review A **96**, 022110 (2017)
arXiv:1704.02994 [quant-ph]

- [P21] Distribution of high-dimensional entanglement via an intra-city free-space link
F. Steinlechner,* S. Ecker,* M. Fink, B. Liu, **J. Bavaresco**, M. Huber, T. Scheidl, and R. Ursin
*These authors contributed equally to this work.
[Nature Communications](#) **8**, 15971 (2017)
[arXiv:1612.00751 \[quant-ph\]](#)
- [P22] Incipient triple point for adsorbed xenon monolayers: Pt(111) versus graphite substrates
A. D. Novaco, L. W. Bruch, and **J. Bavaresco**
[Physical Review B](#) **91**, 161412(R) (2015)
[arXiv:1601.00700 \[cond-mat.mtrl-sci\]](#)

Theses

- [T1] **J. Bavaresco**. Certifying complex quantum properties: High-dimensional entanglement and indefinite causal order. [Doctoral Thesis, Universität Wien \(2021\)](#). Available also on [my website](#).
Examination jury: Andreas Winter, Nicolas Brunner, and Marcus Huber.
- [T2] **J. Bavaresco**. When Bob cannot trust Alice: A semi-device-independent tale of quantum steering. [Master's Thesis, Universidade Federal de Minas Gerais \(2016\)](#). Available also on [my website](#).
Examination jury: Reinaldo Vianna, Rafael Rabelo, and Marcelo Terra Cunha.

Code repositories

- [R1] **J. Bavaresco**, <https://github.com/jessicabavaresco>.
Open-access code repository, containing code pertaining to the research that led to different publications listed above.

Talks, seminars, and lectures

Invited lectures and tutorials

- Sep 2026 **VCQ Summer School 2026** – Vienna Center for Quantum Science and Technology. Vienna, Austria.
“*Quantum vs. Classical correlations*”
(3h course, ~160 registered students)
- Sep 2026 **Qalypso Summer School 2026** – Quantum Optimisation Techniques & Physics of Computation. Valletta, Malta.
“*Semidefinite Programming as an Analytical and Computational Tool for Quantum Information*”
(3h course, ~110 registered students)
- Aug 2026 **QPL 2026 Tutorials** – Quantum Physics and Logic Amsterdam, The Netherlands.
“*Causality in higher-order quantum theory*”
(2h course, ~60 registered students)
- Jul 2026 **La Ricotta Summer School: The Quantum in the Haystack**. Castro Caldelas, Spain.
“*Higher-order quantum transformations: from quantum circuit design to indefinite causal order*”
(7.5h course + 5h tutorial, ~25 registered students)

Invited talks in international conferences

- Sep 2026 **Structure of Quantum theory**. Münster, Germany.
“*Universal Quantum Purification*”

- Feb 2026 **Quantum correlations and measurements.** Les Diablerets, Switzerland.
“The power of catalytic local operations: Activation of Bell nonlocality and quantum state transformations”
- Sep 2025 **HOAQS 2025 – Higher Order Quantum Structures.** Stará Lesná, Slovakia.
“Can the quantum switch be deterministically simulated?”
- Nov 2024 **INAQT 2024 – International Network on Acausal Quantum Technology.** Vienna, Austria.
“Indefinite causal order in boxworld theories”
- Sep 2024 **Causalworlds 2024.** Waterloo, Canada.
“Can the quantum switch be deterministically simulated?”
- Sep 2024 **FQC 2024 – Foundations of Quantum Computing.** London, UK.
“Can the quantum switch be deterministically simulated?”
- Sep 2022 **Quantum Intelligence Workshop 2022.** Birr, Ireland.
“Resource theory of causal connection”

Invited talks in local workshops

- Nov 2025 **IFQIF Workshop.** São Paulo, Brazil. (online)
“Simulating the quantum switch with quantum circuits is computationally hard”
- Apr 2025 **2nd PCQT Workshop – Paris Center for Quantum Technologies.** Paris, France.
“Higher-order quantum computation: Can quantum circuits simulate indefinite causal order?”
- Jan 2025 **TaQC Workshop – Taming Quantum Causality.** Grenoble, France.
“Indefinite causal order in boxworld theories”
- May 2024 **Cryptography in a Quantum World – Paris Rally.** Paris, France.
“Higher-order operations approach to channel discrimination”
- Feb 2024 **Swiss Quantum Days 2024.** Villars-sur-Ollon, Switzerland.
“Higher-order quantum operations and the emergence of indefinite causality”
- Mar 2023 **TOOT – The Order Of Things.** Pisa, Italy.
 Originally invited speaker at a canceled international conference that was to be held in Obergurgl, Austria, in 2020.
“Correlations from boxworld processes”
- Oct 2017 **38. SFB Meeting.** Vienna, Austria.
“Measurements in two bases are sufficient for certifying high-dimensional entanglement”

Contributed talks in international conferences

- Jul 2025 **QPL 2025 – Quantum Physics and Logic.** Varna, Bulgaria.
Plenary talk (single track): *“Indefinite causal order in boxworld theories”*
- Feb 2024 **QUMINOS – Quantum metrology in interacting and open systems.** Les Diablerets, Switzerland.
“Unitary channel discrimination beyond group structures: Advantages of sequential and indefinite-causal-order strategies”

- Aug 2023 **AQIS 2023 – 23rd Asian Quantum Information Science Conference.** Seoul, South Korea.
Long talk (single track): “Unitary channel discrimination beyond group structures: Advantages of sequential and indefinite-causal-order strategies”
- Apr 2021 **YQIS 2021 – VI Int. Conference for Young Quantum Information Scientists.** Michigan, USA.
 “Strict hierarchy between parallel, sequential, and indefinite-causal-order strategies for channel discrimination” (online)
- Jun 2020 **TQC 2020 – 15th Conf. on the Theory of Quantum Comput., Commun. and Crypt.** Riga, Latvia.
 “Semi-device-independent certification of indefinite causal order” (online)
- Aug 2018 **Modern Topics on Quantum Information.** Natal, Brazil.
 “Measurements in two bases are sufficient for certifying high-dimensional entanglement”
- Apr 2018 **CQIS 2018 – Int. Conference for Challenges in Quantum Information Science.** Tokyo, Japan.
 “Measurements in two bases are sufficient for certifying high-dimensional entanglement”

Invited seminars

- Mar 2026 **WOST/QISS Virtual Seminar Series** – International consortium.
- Sep 2025 **Polytechnic Institute of Paris** – Group of Dr. Mirjam Weilenmann. Palaiseau, France.
- May 2025 **University of Toulouse** – Group of Dr. Ion Nechita. Toulouse, France.
- Dec 2024 **The University of Tokyo** – Group of Prof. Mio Murao. Tokyo, Japan.
- Dec 2024 **Hong Kong University (HKU)** – Group of Prof. Giulio Chiribella. Hong Kong.
- Nov 2024 **Université Grenoble Alpes (UGA)** – Group of Dr. Cyril Branciard. Grenoble, France.
- Oct 2024 **Trinity College Dublin** – Group of Prof. Felix Binder. Dublin, Ireland.
- Oct 2024 **Technical University of Vienna (TU Wien)** – Group of Dr. Nicolai Friis. Vienna, Austria.
- Oct 2024 **IQOQI Vienna** – Group of Prof. Ľukáš Brukner. Vienna, Austria.
- Jun 2024 **University of Pisa** – Group of Prof. Costantino Budroni. Pisa, Italy.
- Nov 2023 **University of Bristol** – Group of Prof. Paul Skrzypczyk. Bristol, UK.
- Nov 2023 **Free University of Brussels (ULB)** – Group of Dr. Ognian Oreshkov. Brussels, Belgium.
- Oct 2023 **University of Warsaw** – Group of Prof. Rafal Demkowicz-Dobrzanski. Warsaw, Poland.
- Oct 2023 **Nanyang Technological University (NTU)** – Group of Prof. Mile Gu. Singapore.
- Oct 2023 **Université Grenoble Alpes (UGA)** – Group of Dr. Cyril Branciard (CNRS). Grenoble, France
- Dec 2022 **The University of Tokyo** – Group of Prof. Mio Murao. Tokyo, Japan.
- Nov 2021 **Ulm University** – Group of Prof. Martin Plenio. Ulm, Germany.
- Nov 2021 **National University of Singapore (NUS)** – Group of Prof. Valerio Scarani. Singapore.
- May 2021 **Autonomous University of Barcelona (UAB)** – Group of Prof. Andreas Winter. Barcelona, Spain.
- Mar 2021 **Chapman University** – Group of Prof. Matthew Leifer. Orange, USA.

Jun 2020	S. University of Campinas (UNICAMP) – Group of Prof. Marcelo Terra Cunha. Campinas, Brazil.
Oct 2018	University of Vienna — Group of Prof. Anton Zeilinger. Vienna, Austria
Apr 2018	University of Nagoya – Group of Prof. Francesco Buscemi. Nagoya, Japan.
Apr 2018	The University of Tokyo – Group of Prof. Mio Murao. Tokyo, Japan.
Aug 2017	F. University of Minas Gerais (UFMG) – Group of Prof. Pablo Saldanha. Belo Horizonte, Brazil.
Feb 2017	ICFO – Group of Prof. Antonio Acín. Castelldefels (Barcelona), Spain.

Outreach Activities

Oct 2019	Member of discussion panel and Q&A session with non-STEM students of the course on Methods of Science and Practice in Physics. University of Vienna, Austria.
May 2019	Teaching assistant on one-time lecture to first-year non-physics bachelor students of the course on Quantum Technologies, titled “ <i>What do you mean by random? Randomness, Quantum Mechanics, and Bell Non-locality</i> ”. The University of Tokyo, Japan.
Jul 2018	Press releases concerning our result about high-dimensional entanglement (publication [P17] J. Bavaresco et al., <i>Nature Physics</i> 14, 1032 (2018)) to the general public at Der Standard (Austria) , ORF (Austria) , Salzburger Nachrichten (Austria) , Phys.org (UK) , Catalunya Vanguardista (Spain) , and Innovations Report (Germany) .
May 2018	Invited speaker for an informal talk about quantum physics at a dinner for women interested in STEM topics and living in Vienna, organized by the group “Woman of Vienna STEM”. Vienna, Austria.